CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 4: Cubes and Cube Roots

Every formula and result of the chapter on one sheet

1	Cube	$n^3 = n \times n \times n$
2	Cube of a negative number	$(-a)^3 = -a^3$
3	Cube of a fraction	$\left(\frac{a}{b}\right)^3 = \frac{a^3}{b^3}$
4	Perfect cube test	Prime factors form triplets
5	Cube root	$\sqrt[3]{a^3} = a$
6	Negative number	$\sqrt[3]{-a} = -\sqrt[3]{a}$
7	Product	$\sqrt[3]{ab} = \sqrt[3]{a}\sqrt[3]{b}$
8	Quotient	$\sqrt[3]{\frac{a}{b}} = \frac{\sqrt[3]{a}}{\sqrt[3]{b}}$
9	Sum of cubes	$1^3 + 2^3 + \ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$
10	Zeros	k zeros in n give $3k$ zeros in n^3
11	Edge of a cube	$\sqrt[3]{\text{volume}}$
12	Units digits (number, cube)	2 and 8, 3 and 7, 7 and 3, 8 and 2 swap; others stay